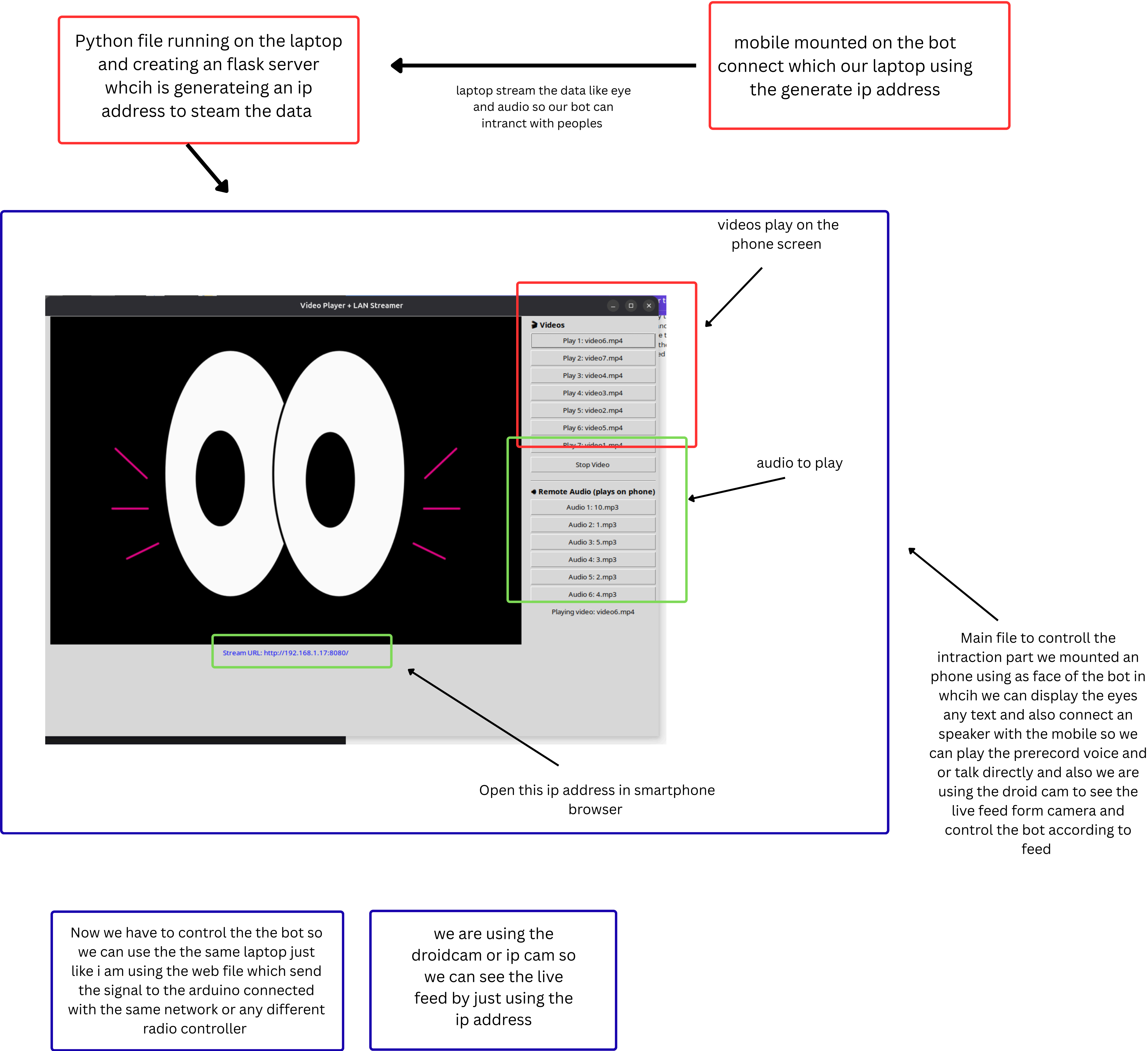


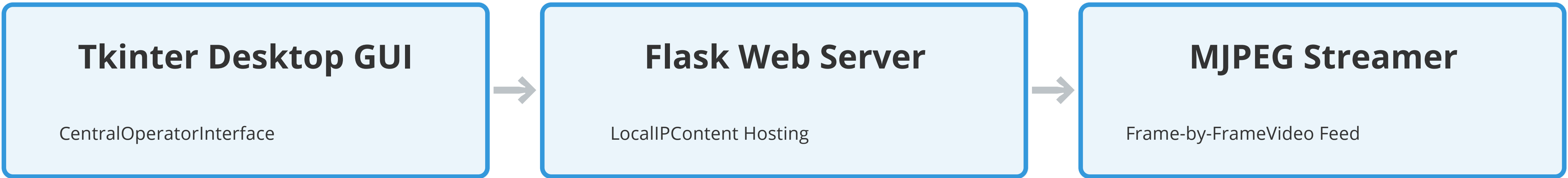
OVERVIEW OF OUR PROJECT

- Our project is simple that we have to create an robot which can intract with the people like talking, show emotion etc but also the project is not too complex we build so we decide make it in simple way.
- WE simply make an bot(four wheel radio or iot control bot) on which we mount our phone so we can intract with the persons using that mobile phone by contolling it with our laptop.
- In our laptop we are running an python file form which we can display any thing on mobile screen and play any sound the python file create an flask server and generate an ip we can simply put that ip our mobile phone and connect with the laptop and play and video and audio
- we can see the person using the front cam of the mobile using droidcam we just simply have to put the droidcam ip address on our laptop browser to see the live feed



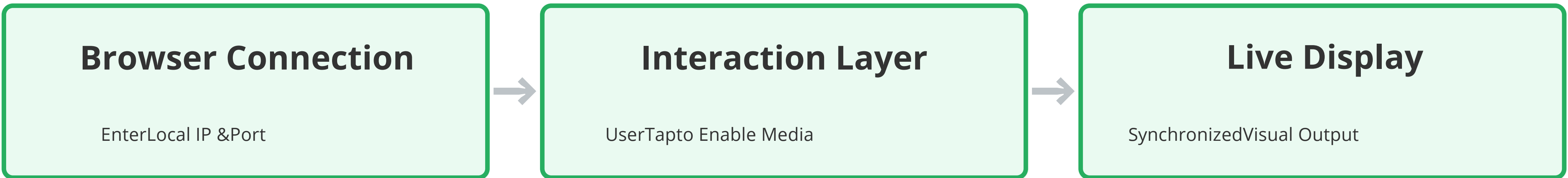
Project Workflow: Interactive Bot - Julie

PHASE 01: HOST SYSTEM (LAPTOP CONTROL)



- **1.1 Initiation:** Launch the Python script to initialize the desktop control window and the backend server.
- **1.2 Video Selection:** Operator selects an animation (.mp4). OpenCV begins extracting frames at the host side.
- **1.3 Network Broadcast:** Flask maps the frames to a local URL (e.g., /video_feed) accessible via the LAN.

PHASE 02: CLIENT SYSTEM (MOBILE BOT FACE)



- **2.1 Handshake:** Mobile device connects to the host IP. The browser requests the MJPEG stream and SSE channel.
- **2.2 Permission Bypass:** User taps the screen. This triggers a "Silent Audio Start" which satisfies browser security policies.
- **2.3 Persistent Feed:** The mobile display maintains a continuous image update loop for smooth eye animations.

PHASE 03: EVENT LOGIC (REMOTE INTERACTION)



- **3.1 Event Dispatch:** When a button is pressed on the Laptop, an SSE (Server-Sent Event) is broadcast to all clients.
- **3.2 Dynamic Loading:** The Mobile JavaScript receives the filename and instantly updates the audio source path.
- **3.3 Real-time Playback:** The audio plays on the mobile speaker in sync with any visual changes on the screen.